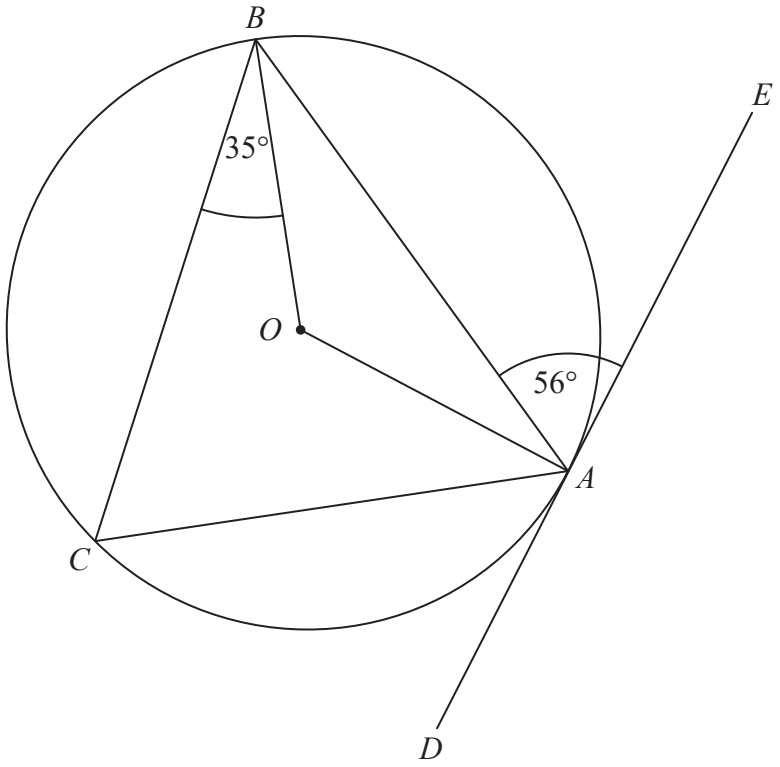
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12



A, *B* and *C* are points on the circumference of a circle, centre *O*.
DAE is the tangent to the circle at *A*.

Angle *BAE* = 56°
Angle *CBO* = 35°

Work out the size of angle *CAO*.
You must show all your working.

OAB = 90° − 56° = 34°

1 mark
Radius is perpendicular to the tangent

OBA = 34°

OA = OB: base angles are equal

ABC = 35° + 34° = 69°

BCA = 56°

1 mark
Alternate segment theorem

BAC = 180° − 69° − 56° = 55°

CAO = 55° − 34° = 21°

1 mark
Subtract OAB from BAC

21 °

(Total for Question 12 is 3 marks)